

When More Data Make the Wrong Model More Certain

Network Dependence, Regularization, and Mean-Field Bias
in Agent-Based SIR Models

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Abstract

Agent-based models can represent heterogeneous agents, local interaction, network structure, and path dependence—but fitting them to coarse aggregate data can produce biased parameter estimates that persist and sharpen as more data are collected at the same observational grain. We demonstrate this using SIR epidemic models [Kermack and McKendrick, 1927]. Data are generated from a network SIR process with known community structure. A mean-field ODE SIR is then calibrated to weekly aggregate incidence counts. The ODE posterior concentrates around a pseudo-true parameter that absorbs network structure into an effective aggregate rate, systematically understating the true mean-field equivalent by 23% in our baseline experiment. Collecting more independent outbreaks at the same coarse grain narrows the posterior fourfold without reducing bias. We connect this result to regularization: the uniform-mixing ODE is the maximally regularized limit of the network model. As community structure strengthens (regularization weakens), ODE bias increases monotonically.

1 The Problem

Agent-based modelers routinely simulate epidemic spread, opinion dynamics, and market behavior at the level of individual agents [Epstein, 2008]. The simulation produces rich internal trajectories: who interacted with whom, in what order, under what conditions. But the data available for calibration are almost always coarser. A modeler fitting an epidemic model to weekly case reports compares a fine-grained stochastic process to aggregate counts measured at a weekly clock.

This mismatch creates a problem that is easy to overlook. When the model is calibrated to coarse data, estimated parameters may be fitting the observation process as much as the underlying mechanism. If the observation process discards information that the mechanism depends on, fitted parameters will not recover the true microscopic rates—no matter how much data are collected.

2 Contribution

We reframe agent-based models as generative statistical models with two components:

$$\text{simulator} \rightarrow \text{latent social process} \rightarrow \text{observation process} \rightarrow \text{data}.$$

The simulator encodes a theory of how agents interact. The observation process encodes how those interactions are measured—at what resolution, with what reporting rate, and with what aggregation structure. A simulation is not yet a statistical model until both components are specified.

3 The SIR Example and the Social Assumption

The standard mean-field ODE SIR model [Kermack and McKendrick, 1927, Keeling and Rohani, 2008] is:

$$\frac{dS}{dt} = -\beta_{ODE} \frac{SI}{N}, \quad \frac{dI}{dt} = \beta_{ODE} \frac{SI}{N} - \gamma I, \quad \frac{dR}{dt} = \gamma I.$$

The term SI is often treated as algebra. It is not. It encodes a strong social assumption: susceptible-infected contact opportunities are proportional to the product of aggregate counts. This is uniform mixing. *The product term is a social assumption.*

4 Network SIR and SI Edges

A network SIR model replaces uniform mixing with an explicit contact structure [Keeling and Rohani, 2008, Kiss et al., 2017]. For susceptible agent i with neighbors $\mathcal{N}(i)$ in contact network A , the infection probability over a short interval follows from the competing-hazards structure of independent Poisson contact processes [Kiss et al., 2017, Ch. 3]:

$$P(i \text{ infected in } [t, t + \Delta t]) = 1 - \exp\left(-\beta \Delta t \sum_{j \in \mathcal{N}(i)} \mathbf{1}\{X_j(t) = I\}\right).$$

Here β is the per-contact, per-infected-neighbor transmission rate—not the same parameter as β_{ODE} . Aggregate incidence depends on $E_{SI}(t)$, the number of susceptible-infected edges [Newman, 2002], not directly on the product SI .

A critical point about scale: β and β_{ODE} are not directly comparable. Under *uniform* mixing with mean degree $\bar{d}$, each susceptible has on average $\bar{d} \cdot I/N$ infected contacts, so the aggregate infection rate is $\beta \cdot \bar{d} \cdot SI/N$. An ODE fitted to data from such a network should therefore recover:

$$\beta_{mf,true} = \beta \cdot \bar{d}.$$

This is not the “true parameter” in a direct sense. It is a *counterfactual reference*: the β_{ODE}

that an unbiased ODE calibration would recover from a network with the same mean degree $\bar{d}$ but no community structure (i.e. Erdős–Rényi). We validate this reference empirically: fitting the ODE to data from an actual uniform network with mean degree $\bar{d} \approx 12$ recovers β_{ODE} within 2% of $\beta_{mf,true}$.

Under community structure, the approximation $E_{SI} \approx cSI$ fails. The epidemic burns rapidly through the seed group, then stalls at group boundaries before spreading slowly across sparse between-group edges. This two-phase dynamics produces an aggregate incidence curve that no single- β_{ODE} ODE can match exactly. The ODE instead concentrates its posterior around a pseudo-true β_{ODE}^* that best approximates the overall pacing of the aggregate curve. The gap $\beta_{ODE}^* - \beta_{mf,true}$ is the bias attributable entirely to community structure.

5 Regularization and the Uniform Graph

We generate contact networks from a stochastic block model [Holland et al., 1983]:

$$\text{logit } P(A_{ij} = 1) = \alpha + \eta B_{ij},$$

where $B_{ij} = 1$ if agents i, j share a group and $B_{ij} = -1/(K - 1)$ otherwise. The parameter η controls community structure; α controls overall density. The regularization penalty is $\lambda\eta^2$.

At $\eta = 0$, all dyads have equal edge probability $\sigma(\alpha)$: the graph is Erdős–Rényi and mixing is uniform. The uniform graph is not the neutral case. It is the maximally regularized case: the network that remains when all group-based distinctions have been shrunk to zero. The standard ODE is correspondingly the maximally regularized aggregate limit. It is not what happens when an ABM gets large. It is what happens when the ABM gets large, coarse, and socially uniform.

6 Simulation Results

Setup. We simulate $N = 1000$ agents in $K = 4$ equal groups. The community network uses $\alpha = -4.4$, $\eta = 2.5$ (mean degree ≈ 37 , within-group edge share ≈ 0.89). True parameters: $\beta = 0.04 \text{ day}^{-1}$ (per contact), $\gamma = 0.10 \text{ day}^{-1}$ (mean infectious period 10 days). The mean-field reference rate is $\beta_{mf,true} = \beta\bar{d} \approx 1.47$. We simulate 150 days, observe weekly aggregate incidence (21 time points), and fit the ODE via grid posterior with γ fixed:

$$\log(1 + Y_k) \sim \mathcal{N}(\log(1 + \hat{Y}_k), \sigma^2).$$

Bias result. The ODE posterior concentrates near $\beta_{ODE}^* \approx 1.13$, well below the counterfactual reference $\beta_{mf,true} \approx 1.47$ —the value the ODE *would* recover from a uniform network of the same density. The bias is -0.33 , or 23% of the reference. The community epidemic’s two-phase shape (rapid within-group spread, slow between-group spread) cannot be reproduced by any

single- β_{ODE} ODE; the posterior instead concentrates on the β_{ODE} that minimizes aggregate misfit. Aggregation can make the ODE look right while making its parameters wrong: the fitted epidemic curve tracks the weekly counts acceptably, but the posterior is centered on a value that has absorbed community structure into a lower effective rate.

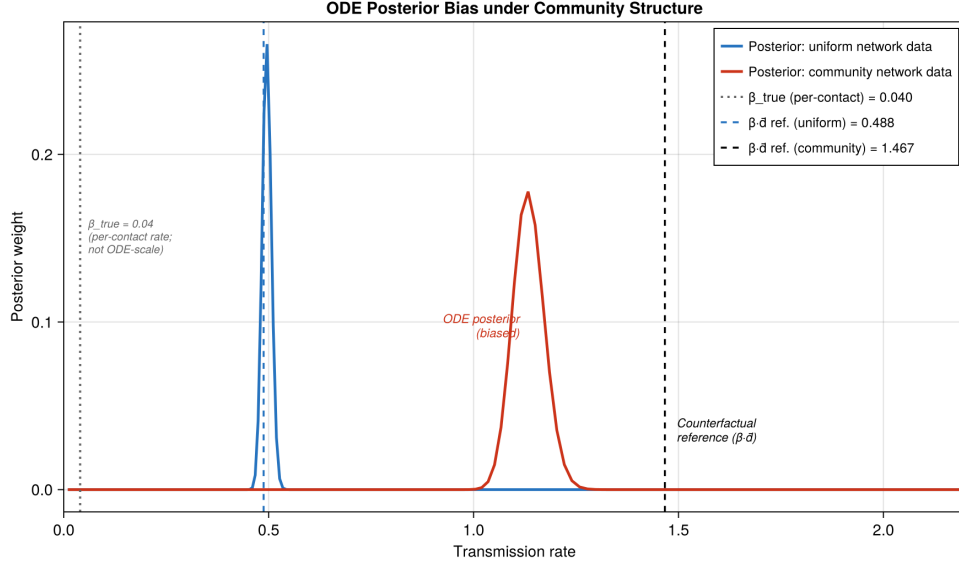


Figure 1: ODE posterior under uniform-network data (blue) sits near its reference $\beta_{mf,true}$; posterior under community-network data (red) concentrates well to the left of its reference (dashed black line), demonstrating systematic bias. Both posteriors are tight—the bias is not uncertainty, it is misspecification.

More data, same grain. We generate R independent community-network outbreaks and pool their likelihoods. As R increases from 1 to 20, the posterior 95% credible interval width shrinks from 0.136 to 0.036—a fourfold reduction. The posterior MAP stays in the range $[1.11, 1.22]$, always well below $\beta_{mf,true} \approx 1.47$.

More data at the same observational grain do not recover the missing mechanism. They estimate the wrong reduced model more precisely. Variance shrinks. Bias remains.

Regularization path. Varying η from 0 to 3 and pooling 20 independent outbreaks per level to obtain stable pseudo-true estimates, ODE bias increases strictly monotonically: -0.044 (9% of reference) at $\eta = 0$ and -0.48 (23% of reference) at $\eta = 3$. The relationship is convex: bias is modest for small η and accelerates as community structure becomes strong. As community structure strengthens, the epidemic increasingly spreads in group-wise waves; the ODE compensates by fitting a lower effective transmission rate.

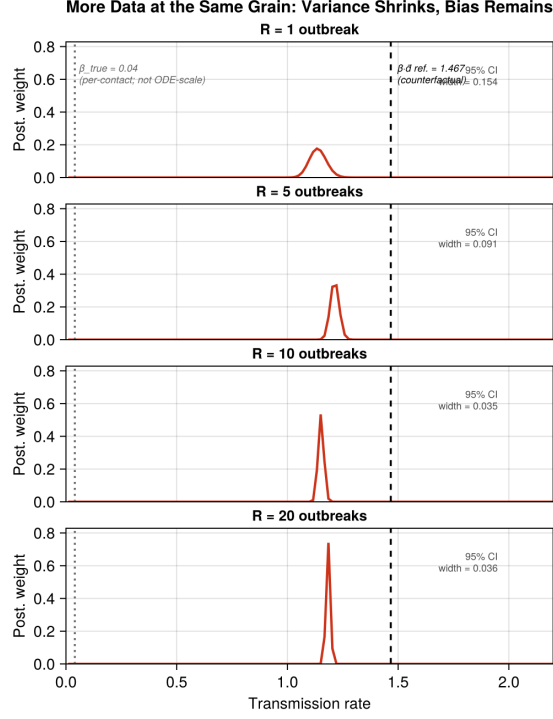


Figure 2: Four stacked panels showing the ODE posterior for β_{ODE} as the number of independent community-network outbreaks increases ($R = 1, 5, 10, 20$, top to bottom). All panels share the same horizontal axis. The dotted vertical line marks $\beta_{true} = 0.04$ (per-contact rate; a different quantity than β_{ODE}). The dashed vertical line marks the counterfactual reference $\beta_{mf,true} \approx 1.47$ (what an unbiased ODE would recover from a uniform network of the same density). The 95% credible interval width, annotated in each panel, shrinks fourfold from $R = 1$ to $R = 20$; the posterior mass does not move toward the reference. More data make the wrong model more certain.

7 Robustness and Model Criticism

This connects to PAC-Bayesian ideas [McAllester, 1999]: a model must pay for how far it moves from prior plausible mechanisms in order to fit the data. More generally, regularization demands stability under modeling changes the theory regards as irrelevant. Findings that survive perturbations across network realizations, observation grains, and community structure strengths are the robust ones.

A practical workflow follows: (1) specify the latent process and observation process separately; (2) compare richer simulations against their mean-field limits; (3) penalize unnecessary heterogeneity in agents and network structure; (4) vary observation grain and clock; (5) examine posterior bias; (6) run posterior predictive checks.

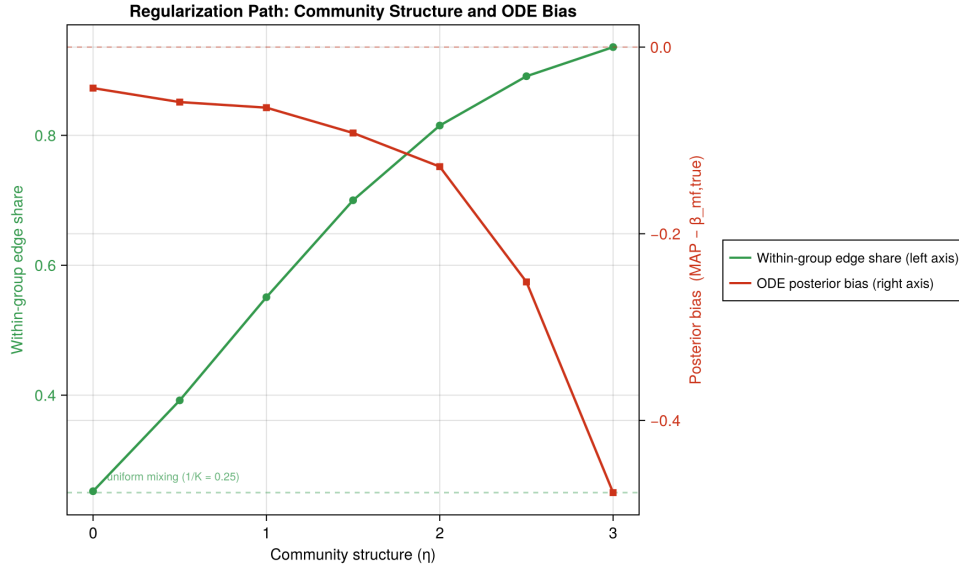


Figure 3: Regularization path. As community structure η increases (green, left axis), the within-group edge share rises from uniform mixing ($1/K = 0.25$) toward 1. ODE posterior bias (red, right axis) decreases monotonically and convexly. At $\eta = 0$ the network is approximately Erdős–Rényi and bias is near zero; at $\eta = 3$ bias reaches -0.48 .

8 Conclusion

Fitting a mean-field ODE to community-network epidemic data produces systematically biased parameter estimates. More data at the same coarse grain sharpens the bias rather than correcting it. The analyst becomes more confident—but about the wrong model. The broader goal is to make social simulation empirically disciplined: not merely to generate plausible patterns, but to determine which mechanisms are needed, which can be regularized away, and where a model’s claims cease to be robust.

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